Optimal Composition of Sovereign Debt: An Adaptation from Modern Portfolio Theory

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ABSTRACT

Finding the optimal composition of the sovereign debt and the relevant borrowing strategies has vital value for debt managers as accurate strategies would help saving a significant amount of taxpayers' money. This is crucial especially when the level of debt is high. In that regard, exploring the optimal composition and applying targeted policies could contribute to efficiency and effectiveness of debt management policies. Even though converting the current liability portfolio swiftly in order to reach an optimal composition in the short-term would be infeasible due to borrowing constraints in some financial markets, yet it would be an important tool paving the way for approaching a more cost effective composition.

In this paper, we used these borrowing constraints in order to reduce the cost/risk calculation burden of numerous strategy alternatives. That is, setting lower and upper bounds for the shares of borrowing instruments eliminated many infeasible strategies and facilitated reducing the time spent in calculations substantially. Provided that a sovereign has the main requisites of a simulation module, financing program and macroeconomic forecasts, finding the optimal composition could be a straightforward process. Therefore, we believe that it would be a very efficient and clear way to find the optimal composition of debt stock for sovereigns which have several constraints on borrowing options.

JEL Classification: H63, G11, C63
Keywords: optimal debt composition, sovereign debt management, optimal borrowing strategies, simulation modeling

1. Introduction

Public debt managers meet public financing needs by raising funds in both domestic and external markets. These funding often consist of selling fixed income securities, at various maturities, currencies and interest payment structures. The goal of debt managers is to borrow at lowest possible cost, by taking into account risk preferences. Since there is a trade-off between cost and risk, less costly borrowing options usually bring along more risk.

One of the main risks faced by public debt portfolios relate to market risk. Market risk refers to the risk of increases in the cost of the debt arising from changes in market variables, such as interest rates and exchange rates (IMF, 2016). To manage market risks, debt managers use deterministic and stochastic decision-making methods. A widely popular metric to measure market risk is "Value-at-

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Risk", which is generally used by banks and other financial institutions. "Value-at-Risk" is used to control level of risk exposure of a portfolio, that is, extent of possible losses in market value of investments.

In debt management perspective, the focus is not the decrease in market value, but the increase in cost of debt. Therefore, in debt management offices variations of Value-at-Risk are used to control risks, which are "Cost-at-Risk" and "Cash Flow-at-Risk" models. In this context, debt managers need to choose a borrowing strategy based on their risk preferences. Even though there is a bunch of research on finding the optimal composition of sovereign debt on theoretical grounds, the usefulness of these findings in practice is open to debate as the complexity of these methodologies makes them very difficult for high level decision makers to be adopted. In this paper, we propose a plain method to find the optimal composition of public debt derived from a relevant borrowing strategy. To apply the method presented in this paper, the risk managers only should have three modules at their disposal, which can be summarized as follows:

- ***A simulation module*** either stochastic or deterministic, which gives cost and risk characteristics of different borrowing strategies at the end of simulation horizon,
- ***A financing program*** which calculates borrowing need and
- ***A debt stock database*** to be used in simulation module, and a set of macroeconomic projections.

Against this background, the rest of the paper is organized as follows. Section 2 describes the simulation model used by Turkish Treasury for Debt Strategy Analysis, which is the starting point of the procedure of finding the optimal composition. Section 3 describes the method in two stages. First, an efficient frontier is determined among some possible set of borrowing strategies. This set of strategies is specified considering borrowing constraints of Turkey, which can also be constraints of other developing countries. Then a relevant strategy is determined from this efficient frontier. This section also addresses how we reached optimal composition by using the efficient frontier. Finally, Section 4 provides a brief conclusion and main findings of the paper.

2. Turkish Treasury Simulation Model for Debt Strategy Analysis

Turkish Treasury Debt Simulation Model (TDSM) is a stochastic model to assist the borrowing strategy decision making process in Turkish Treasury. TDSM is an in-house built, Matlab based model which has been used since 2003 with several improvements. Briefly stated, it takes initial debt stock, non-borrowing financing sources (i.e. primary balance, change in cash balance, privatization revenues etc.) historic macroeconomic data and borrowing strategies to be tested as inputs. It first simulates macroeconomic variables, then calculates borrowing need and produces related statistics of different borrowing strategies at the end of 5-year horizon as output.

First, the process starts with reading debt stock data, which includes all required information[3] about debt stock. Borrowing requirement stems from service of existing debt, service of future debt (debt issued in simulation horizon) and deficits in non-borrowing items stated above. Calculation of borrowing requirements arises from existing debt stock is fed by this database.

[3] Debt stock database contains following attributes of each issue: instrument type, currency type, amount, value date, maturity date, coupon rate, spread, yield to investor, coupon period and next coupon date.

Second, a financial scenario generator creates projections of interest rates, exchange rates, and consumer price indexes for the 5-year simulation period. A longer horizon could be able to better capture roll-over costs of the bonds issued in the initial years. However, as the simulation horizon lengthens, reliability of macroeconomic variable forecasts become more of an issue. Additionally, it is self-evident that longer horizon results in an increase in time spent in computations. In that regard, considering those pros and cons we kept the simulation period at 5 years.

In the simulation module, short-term interest rates are modelled by using Cox, Ingersoll and Ross (CIR, 1985) and Chan, Karolyi, Longstaff and Sanders (CKLS, 1992) models both with mean reversion properties. To overcome CIR's weakness in predicting long-term interest rates, a hybrid model with both mean reversion and dynamic term-structure fitting features is used to estimate and forecast yield curves for TRY, USD and EUR. As for the term structure part of the hybrid model, methodological approach presented by Diebold and Li (2006) is used.

Exchange rates and price indexes are modelled via geometric Brownian motion, which is a stochastic process based on random walk. Financial scenario simulation also incorporates correlation between macroeconomic variables. The correlation is created by decomposing the covariance matrix using Cholesky decomposition.

TDSM uses the Conditional Cost-at-Risk metric (also sometimes referred to as the expected tail loss or expected shortfall) which is an extended version of cost-at-risk. It focuses on the average of expected cost values in the worst-case scenarios which occur beyond the specified confidence level, rather than a single value on a percentile. This metric enables the user to capture tail risks and includes scenarios beyond confidence interval into decision making process.

Lastly, a cash-flow engine calculates risk and cost metrics of different borrowing strategies. This engine initializes itself with the debt database and borrowing requirements, loops over stochastically generated macroeconomic scenarios, issues new debt and redeems maturing debt. More details about Turkish Treasury Debt Simulation Model can be found in Balibek and Memis (2012).

3. An Approach to Find the Optimal Composition of Sovereign Debt

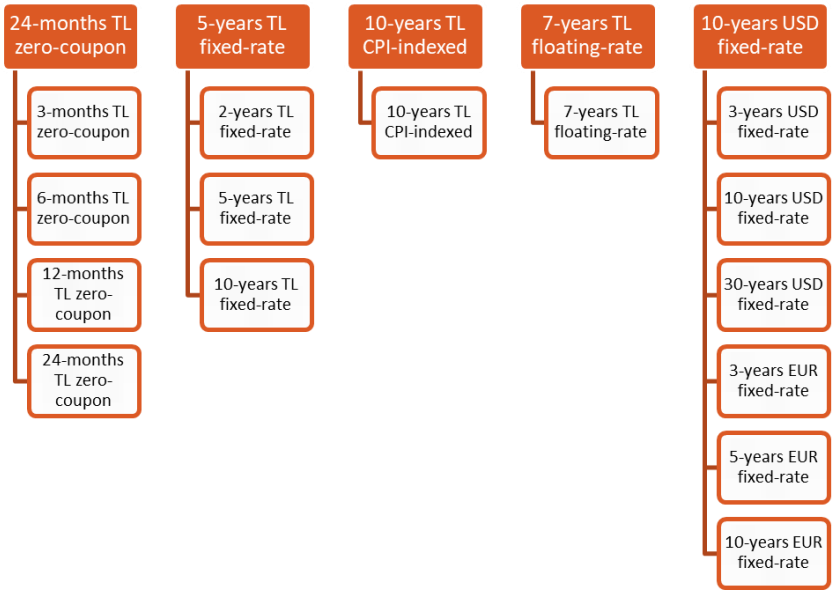
Our approach aims to enable public debt management experts to utilize their borrowing strategy simulation modules in determining optimal debt composition, without requiring any other optimization program. The approach presented here is based on finding a relevant borrowing strategy that lets us reach the optimal composition of sovereign debt. Therefore, first we need to find an "optimal" borrowing strategy.

An optimal borrowing strategy could be found by employing non-linear optimization techniques. However, these techniques increase complexity of the process, which may slow down its implementation and may impede communication with top management. An approach to find optimal borrowing strategy is proposed by Bolder and Rubin (2007). In their work, they underline the computational challenge of optimizing a high-dimensional non-linear function and address the issue approximating the objective function by using function approximation techniques and then to optimize this approximation. Even though, this is a reasonable solution to the problem, the methodology must be simple enough to be communicated to the top level decision-makers.

To deal with computational expenses, complexity of optimization models and the comprehensibility issues, we propose another solution for finding the optimal debt composition. Also, we imposed some simplifying adjustments on borrowing strategies. To determine possible borrowing strategies, we set

the granularity of borrowing to be 10% of total borrowing need. All possible strategies are generated via a simple VBA program. Given there are 15 possible debt instruments in Turkish Treasury Simulation Model, we determined 5 debt instruments, which could possess certain characteristics to represent a certain group of instruments (i.e. a single type of USD-denominated instrument is used to represent instruments in foreign currency). The representative instruments are provided in Table 3.1.

Table 3.1: Representative Instruments



Source: Authors' calculations

After finding the representative instruments, by considering risk characteristics of specific types of securities and using judgement about borrowing markets, we set lower and upper bounds to the issuance shares of securities. While doing so, to see the individual risk characteristics of securities we followed a method, in which all borrowing need in a strategy is met by a single security type and we tested 5 strategies. In our case, in spite of limited cost advantage, FX borrowing has the highest risk. On the other hand, highest cost expectation comes from a 100% CPI-indexed borrowing strategy.

Because of high exchange rate risk, we set 30% limit to FX borrowing and 60% limit to other TL borrowing alternatives. We set a zero lower bound to FX borrowing. On the other hand, as exclusion of a specific type of TL borrowing instrument usually affects the costs of other instruments negatively, we set 5% lower bound to TL securities. Upper and lower bounds can be seen in Table-3.2.

Table 3.2: Issuance Limits of Representative Instruments

	Lower Bound	Upper Bound
24-months TL zero-coupon	5%	60%
5-years TL fixed-rate	5%	60%
10-years CPI-indexed TL	5%	60%
7-years floating-rate TL	5%	60%
10-years fixed-rate USD	0%	30%

Source: Authors ' calculations

Our work comprises of two stages. First, an efficient frontier is formed among all tested strategies. Secondly, the relevant optimal strategy is chosen from the efficient frontier.

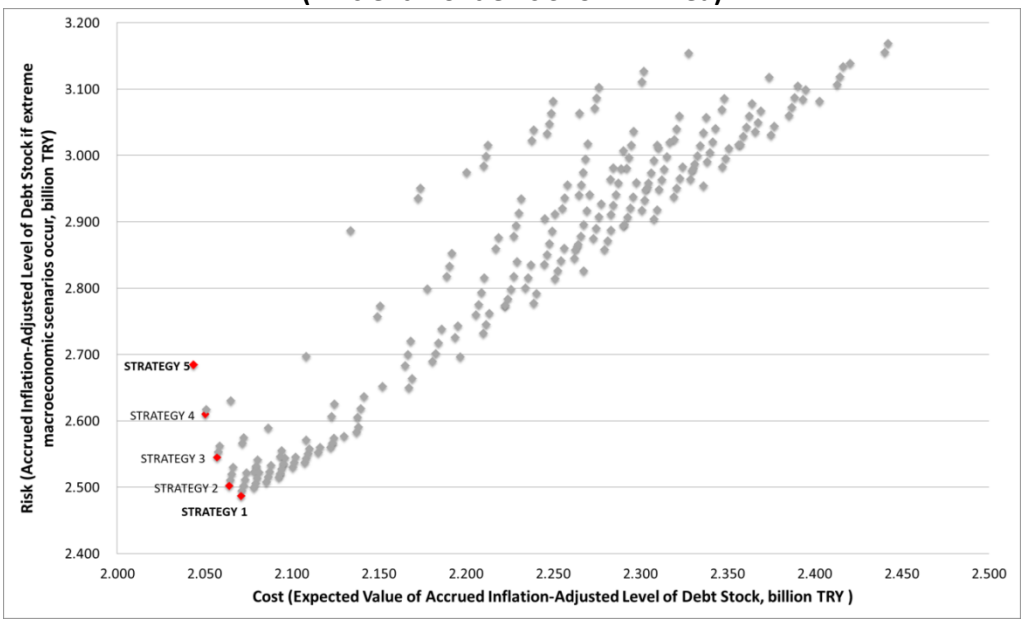
3.1. Finding the Efficient Frontier

To find efficient frontier, under 5-years simulation horizon, we started by testing cost and risk of 191 different borrowing strategies with 10% granularity. After that, the granularity was reduced to 5% and 53 additional strategies were tested. Evaluating 244 strategies with 5000 randomly generated scenarios took 9 hours 26 minutes for a 2016-made computer[4]. If we had not implemented simplifying adjustments mentioned in the previous section, computation time would increase exponentially to unfeasible levels. For instance, setting the granularity for the share of an instrument at 1% requires billions of strategies to be tested. This implies decades-long calculation time and will make it impossible to solve the problem by applying the simple method presented in this paper.

To capture optimal debt stock composition through the relevant borrowing strategy, we started with a state where there is zero debt stock and we kept each strategy unchanged during the course of simulation period. Also, we kept borrowing needs during simulation period same as what real debt stock and mean macroeconomic projections imply. Thus, risk and cost levels of different strategies were not affected by initial zero-stock state. In that regard, starting with zero-stock state enabled us to reveal optimal debt stock composition by using relevant borrowing strategy.

As cost indicator we used the value of "Accrued Inflation-Adjusted Debt Stock" at the end of 5-year analysis horizon. This metric captures accrued interest of CPI-linked bonds and effects of currency fluctuations on stock. Extreme values of these stock figures at the end of analysis period are regarded as the risk indicators. Extreme values occur when extreme macroeconomic scenarios occur. In our work, we aimed optimizing this metric. Figure 3.1 provides the results of 244 different borrowing strategies.

Figure 3.1: Cost-Risk Chart of 244 Different Borrowing Strategies (Efficient Frontier is Shown in Red)



Source: Authors' calculations

[4] Processor: Intel® Core™ i5-6500 CPU 3.20 GHz, x64 based processor. RAM: 8GB.

A visual analysis of Figure 3.1 exhibits that there are some strategies that drag both cost and risk in positive direction. This situation stem from some instruments that have both high cost and risk values. In fact, this methodology can be used to find whether there are any such questionable securities exist in treasury's toolkit. Even though these controversial securities deteriorate the debt portfolio from a cost/risk perspective and signal the debt managers not to use these instruments at all, demand dynamics in the borrowing market might force debt managers to issue certain amount of these instruments. This is a factor, which we need to consider while setting lower bounds to TL instruments. In addition, high correlation between costs of borrowing instruments limits the diversification gains and forces certain strategies containing certain instruments to stand out. To grasp this effect better, please see the compositions of efficient strategies presented in Table 3.3.

Table 3.3: Compositions of Efficient Strategies

Strategies (from lowest to highest risk)	Zero coupon	Fixed-rate coupon	CPI-indexed	FRN	FX-denominated
1	60%	25%	5%	5%	5%
2	60%	5%	15%	20%	0%
3	60%	15%	10%	15%	0%
4	60%	15%	10%	5%	10%
5	60%	5%	5%	5%	25%

Source: Authors' calculations

Instead of a single instrument type (zero-coupon in this case) be prominent, we would normally have expected a more diversified portfolio weights because of diversification gains. However, high correlation of costs between instruments that are all issued by a single issuer, almost cancels out diversification gains. Therefore, an instrument which has slightly better cost and risk characteristics dominates those efficient strategies. Moving from the lowest risk strategy (Strategy 1) to the lowest cost strategy (Strategy 5), an increase in share of FX-denominated instrument is apparent. The strongest cost and risk trade-off occurs between FX and TL instruments. This is because USD 10 year instrument has the lowest correlation with each other instrument. In that regard, new instruments such as a GDP-linked bond here may enhance risk and cost metrics of portfolios because of its lower correlation with other instruments (Yavuz, 2017). Thus, as the introduction of such instruments would change the correlations between instruments, it would lead to notable changes in the efficient frontier (Figure 3.1) and efficient strategies (Table 3.3).

Table 3.4: Correlation Matrix (Jan 2010- Sep 2018)

	Zero2Y	Fixed5Y	CPI	FRN	USD10Y
Zero2Y	1.00	0.98	0.82	0.88	0.75
Fixed5Y	0.98	1.00	0.82	0.87	0.75
FRN	0.88	0.87	0.80	1.00	0.55
CPI	0.82	0.82	1.00	0.80	0.57
USD10Y	0.75	0.75	0.57	0.55	1.00

Source: Authors' own calculations

3.2. Selecting the Relevant Strategy from the Efficient Frontier

Modern Portfolio Theory, pioneered by Markowitz (1952), proposes an approach to reach optimal portfolio composition. But, this approach is to determine optimal asset composition for investors

holding different assets. We therefore modified this approach considering borrower position of sovereigns. Details can be found in the Appendix.

First, for each point on the efficient frontier, we calculate the delta of risk value (marginal conditional cost-at-risk (MCCaR)[5]) and the additional cost occurred by the issuance of one additional unit of that instrument for each instrument type. Then, we calculated the marginal risk/marginal cost ratios for each instrument type. After that, among the points on the efficient frontier, the one in which marginal risk/marginal cost ratio is the closest to each other is regarded as the optimal portfolio. This states that, at their current proportions in the portfolio, the trade-off between the cost and risk is equivalent for all instruments and therefore the portfolio is balanced in an optimal way in terms of marginal cost and marginal risk.

$$\frac{\text{MCCaR}_{p1,i1}}{\text{Marginal Cost}_{p1,i1}} = \frac{\text{MCCaR}_{p1,i2}}{\text{Marginal Cost}_{p1,i2}} = \cdots = \frac{\text{MCCaR}_{p1,i5}}{\text{Marginal Cost}_{p1,i5}}$$

$$\frac{\text{MCCaR}_{p2,i1}}{\text{Marginal Cost}_{p2,i1}} = \frac{\text{MCCaR}_{p2,i2}}{\text{Marginal Cost}_{p2,i2}} = \cdots = \frac{\text{MCCaR}_{p2,i5}}{\text{Marginal Cost}_{p2,i5}}$$

$$\cdots \quad = \quad \cdots \quad = \quad \cdots$$

$$\frac{\text{MCCaR}_{p5,i1}}{\text{Marginal Cost}_{p5,i1}} = \frac{\text{MCCaR}_{p5,i2}}{\text{Marginal Cost}_{p5,i2}} = \cdots = \frac{\text{MCCaR}_{p5,i5}}{\text{Marginal Cost}_{p5,i5}}$$

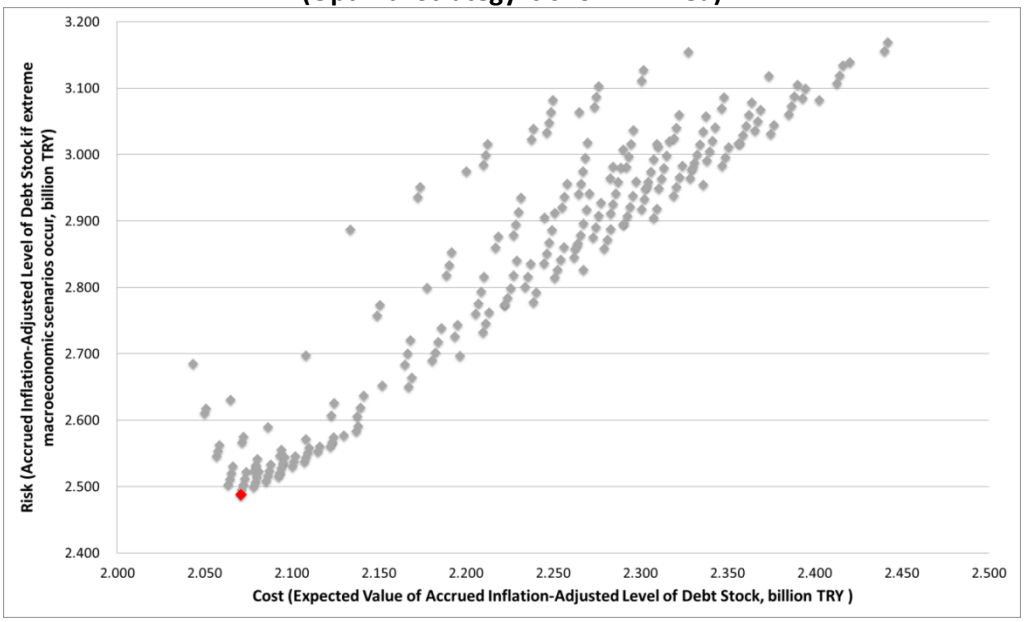
where p = 1…5 denotes portfolios on efficient frontier, i = 1…5 denotes instrument types in each portfolio.

We checked the condition above by comparing ranges, standard deviations and mean absolute deviations of each strategy. According to the calculations, those three metrics are minimum in the composition which implies minimum risk (Strategy-1 in Figure 3.2). Therefore this composition is also determined as the optimal composition. The composition consists of 60% zero-coupon, 25% fixed-rate coupon, 5% floating-rate, 5% CPI-indexed TL denominated bonds and 5% foreign currency denominated bonds. This composition enables the total risk to be reduced by 13 percent and the cost by 9 percent, in comparison to the composition in which all instrument types have equal shares. This amount, of course, can vary depending on the current composition and the strategy that is planned to be applied, yet considering a level of debt stock around USD 200 billion; even the slightest reduction might lead to considerable cost savings.

An important point to consider here is that the macroeconomic assumptions which are the starting point of the weights that lead to the optimal composition and the simulation values obtained based on these assumptions may change over time. Therefore, these assumptions are critical when moving to the optimal composition.

[5] MCCaR can be interpreted as a change in the CCaR for a $1 or 1% change in the position. Although that is not strictly correct, it is a reasonable approximation of the concept behind marginal CCaR, which is to reflect the impact of a small change. In a diversified portfolio, marginal CCaR may be used to determine the contribution of each borrowing instrument to the overall CCaR and the marginal CCaRs for all positions may be proportionately weighted to sum to the total CCaR.

Figure 3.2: Cost-Risk Chart of 244 Different Borrowing Strategies (Optimal Strategy is shown in Red)



Source: Authors' own calculations

4. Conclusions

Sovereign debt managers have the duty of providing the necessary funding for the public expenses in a timely and consistent manner. For that reason, debt management is a continuous process that needs utmost care. In many countries, the public debt stock stands as the biggest liability portfolio in the country. Therefore, even minor improvements regarding the composition of the stock and the borrowing strategies could have major impacts on the cost of debt financing and accordingly the welfare of the citizens. However, it is not always the case that the debt managers have the possibility to use the borrowing instruments in desired size, maturity, level, currency and interest type.

In most of the emerging markets, the development level of the borrowing markets draws the limits for the diversity of borrowing instruments. Those limitations also oblige the use of several instruments more than the desired levels, causing a sub-optimal composition of debt stock and borrowing. Yet, some other factors such as the desire to diversify the sources of funding, reaching out to new investors, providing a variety of investment tools to investors and supporting national reserves might induce the use of instruments, which are not optimal from a pure cost-risk perspective. Such instruments may not directly contribute to the debt management, but they might support other targets such as playing a benchmark role for the corporates, sectoral development and reserve management.

Against this backdrop, we proposed a simple approach to find the optimal composition of debt and the relevant borrowing strategies to reach such composition. The sole requirements are a simulation module, a financing program and a debt stock database. By using these three tools, the user needs to decide the granularity and the upper-lower bounds for the strategies that need to be tested. However, it is important to underline that the models always produce output based on the inputs provided. In that regard, the macroeconomic assumptions play a key role for attaining credible results. Moreover, such assumptions might change over time. For that reason, frequencies of such analysis and calibration procedures for the portfolio are important issues that need to be considered as well.

Nowadays, in the era of advanced technologies we believe that computational needs are not a burden anymore. For instance, in our case with 10% granularity and 5 tenors, we got the simulation results for 244 different borrowing strategies in less than 10 hours even though our models and modules are not optimized for computational efficiency. Such speed could also enable daily calibration and use of active debt management tools, once the optimal structure is obtained.

All in all, the methodology presented in this paper provides a simple and coherent way for practitioners who would like to improve the composition of the debt stock and would like to discover the relevant borrowing strategies for more efficient management of debt.

APPENDIX: Method Used in Finding Optimal Debt Portfolio

According to approaches based on Modern Portfolio Theory, optimal portfolio is defined as the composition, in which the ratio of excess return divided by the per unit change in the risk value of the portfolio when an additional investment of that position is added to the portfolio, is equal to the same constant for each position. In other words, optimal composition is obtained when this ratio for a position is equal to the ratios of other positions.

To recall, Sharpe ratio is a ratio shows the return of an instrument above risk-free return (excess return) per unit of standard deviation of the portfolio.

$$\text{Sharpe ratio} = \frac{Portfolio\ Return - Riskfree\ Return}{Portfolio\ Standard\ Deviation} \quad (1)$$

Value-at-Risk is defined as the value that can be expected to be lost during severe adverse market fluctuations. (Marisson, 2002) If we take Value-at-Risk instead of standard deviation as risk indicator, the ratio turns into as equation (2).

$$\frac{Portfolio\ Return - Riskfree\ Return}{Portfolio\ Value\text{-}at\text{-}Risk\ (VaR)} \quad (2)$$

This ratio maximizes when the weights of each position makes the equality below hold for each position or instrument type.

$$\frac{\text{Return}_{\,instrument\ i} - Riskfree\ Return}{(\text{Marginal Value-at-Risk}_i)} = \frac{\text{Return}_{instrument\ j} - Riskfree\ Return}{(\text{Marginal Value-at-Risk}_j)} \quad (3)$$

Since there is no "return" and "risk-free return" notions for sovereign Treasuries, we looked at how an additional issuance of an instrument changes the risk taken and the cost borne relative to other instruments. Therefore, considering sovereign borrower position, the general equation in question is modified and applied as follows:

$$\frac{\text{MCCaR}_{1,1}}{\text{Marginal Cost}_{1,i1}} = \frac{\text{MCCaR}_{1,2}}{\text{Marginal Cost}_{1,2}} = \cdots = \frac{\text{MCCaR}_{1,I}}{\text{Marginal Cost}_{1,I}}$$

$$\frac{\text{MCCaR}_{2,1}}{\text{Marginal Cost}_{2,1}} = \frac{\text{MCCaR}_{2,2}}{\text{Marginal Cost}_{2,2}} = \cdots = \frac{\text{MCCaR}_{2,I}}{\text{Marginal Cost}_{2,I}}$$

$$\ldots \quad = \quad \ldots \quad = \quad \ldots$$

$$\frac{\text{MCCaR}_{P,1}}{\text{Marginal Cost}_{P,1}} = \frac{\text{MCCaR}_{\boldsymbol{P},2}}{\text{Marginal Cost}_{\boldsymbol{P},2}} = \ldots = \frac{\text{MCCaR}_{\boldsymbol{P,I}}}{\text{Marginal Cost}_{\boldsymbol{P,I}}}$$

Turkish Treasury adopted "Conditional Cost-at-Risk" method to compare alternative borrowing strategies and to determine a benchmark borrowing policy. According to this method, the average of cost values, which are expected to occur in the tail of the cost distribution, is accepted as risk measures. The cost distribution is obtained via Monte Carlo simulations with 5000 iterations. Occurrence of those tail scenarios depend on the occurrence of tail scenarios in financial scenario generation module (i.e. macroeconomic scenario simulations).

In Marginal Cost and Marginal Risk calculations, one percent of total borrowing requirement is set to be one unit of additional issuance. However, to isolate the effect of this issuance to risk and cost measures, this additional issuance is not meet by decreasing the issuance of other debt instruments but by increasing the borrowing 101 percent of borrowing requirement. Since Marginal Risk/Marginal Cost ratio is examined only within a strategy for each type of instruments in that strategy, 101 percent borrowing has no effect on comparing between different borrowing strategies.

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